

TOTIENT FIBRE EXTREMES

GPT-5.5 PRO

ABSTRACT. Let $f_{\max}(n)$ and $f_{\min}(n)$ denote respectively the largest and smallest integers m with $\phi(m) = n$, whenever such integers exist. We determine the maximal possible separation between the largest and smallest elements of a totient fibre. More precisely, if the maximum is taken over totient values $n \leq x$, then

$$\max_{\substack{n \leq x \\ n \in \phi(\mathbb{N})}} \frac{f_{\max}(n)}{f_{\min}(n)} = (e^\gamma + o(1)) \log \log x.$$

The upper bound follows from the extremal order of $m/\phi(m)$, while the lower bound is obtained by a construction using Linnik's theorem on the least prime in an arithmetic progression. We also record a simple permanence observation showing that any one nontrivial totient collision propagates to infinitely many.

1. PRELIMINARIES

Throughout, ϕ denotes Euler's totient function and γ denotes Euler's constant. Since $f_{\min}(n)$ and $f_{\max}(n)$ are defined only when n is a totient value, we interpret the expression in the problem as

$$\mathcal{R}(x) := \max_{\substack{n \leq x \\ n \in \phi(\mathbb{N})}} \frac{f_{\max}(n)}{f_{\min}(n)}.$$

The maximum is nonempty for $x \geq 1$, since $1 = \phi(1) = \phi(2)$.

We first note that every totient fibre is finite. Indeed, for an odd prime power p^a ,

$$\phi(p^a)^2 = p^{2a-2}(p-1)^2 \geq p^a,$$

and for 2^a ,

$$\phi(2^a)^2 = 2^{2a-2} \geq 2^{a-1}.$$

Multiplying over prime powers gives

$$\phi(m)^2 \geq \frac{m}{2}$$

for every $m \geq 1$. Hence if $\phi(m) = n$, then

$$m \leq 2n^2.$$

Thus $f_{\min}(n)$ and $f_{\max}(n)$ are genuine finite quantities whenever $n \in \phi(\mathbb{N})$.

We shall use the following standard unconditional estimates: the prime number theorem in the form

$$\vartheta(y) := \sum_{p \leq y} \log p \sim y,$$

Mertens' product theorem

$$\prod_{p \leq y} \left(1 - \frac{1}{p}\right)^{-1} \sim e^\gamma \log y,$$

and Linnik's theorem, which says that there are absolute constants $C, L > 0$ such that, for every integer $a \geq 1$, the least prime $\ell \equiv 1 \pmod{a}$ satisfies

$$\ell \leq Ca^L.$$

Increasing L if necessary, we may assume $L \geq 1$.

We shall also need the following familiar consequence of the prime number theorem and Mertens' theorem.

Lemma 1.1. *As $T \rightarrow \infty$,*

$$\max_{1 \leq m \leq T} \frac{m}{\phi(m)} = (e^\gamma + o(1)) \log \log T.$$

Proof. Let $p_1 < p_2 < \dots$ be the sequence of primes and put

$$N_k := \prod_{i=1}^k p_i, \quad N_0 := 1.$$

Choose $k = k(T)$ so that

$$N_k \leq T < N_{k+1}.$$

If $m \leq T$ and $r = \omega(m)$, then

$$N_r \leq \text{rad}(m) \leq m \leq T,$$

so $r \leq k$. Since the function $p \mapsto p/(p-1)$ is decreasing on the primes,

$$\frac{m}{\phi(m)} = \prod_{p|m} \frac{p}{p-1} \leq \prod_{i=1}^r \frac{p_i}{p_i-1} \leq \prod_{i=1}^k \frac{p_i}{p_i-1}.$$

Equality is attained at $m = N_k$. Hence

$$\max_{1 \leq m \leq T} \frac{m}{\phi(m)} = \prod_{i=1}^k \frac{p_i}{p_i-1}.$$

By Mertens' theorem,

$$\prod_{i=1}^k \frac{p_i}{p_i-1} = (e^\gamma + o(1)) \log p_k.$$

On the other hand,

$$\log N_k = \vartheta(p_k) \sim p_k,$$

and since $T < N_{k+1}$,

$$\log T < \log N_{k+1} = \vartheta(p_{k+1}) \sim p_{k+1} \sim p_k.$$

Together with $\log T \geq \log N_k \sim p_k$, this gives $p_k \sim \log T$. Therefore

$$\log p_k \sim \log \log T,$$

and the lemma follows. □

2. THE ASYMPTOTIC FORMULA

Theorem 2.1. *As $x \rightarrow \infty$,*

$$\mathcal{R}(x) = (e^\gamma + o(1)) \log \log x.$$

Proof. We prove the upper and lower bounds separately.

First let $n \leq x$ be a totient value, and put

$$M = f_{\max}(n), \quad m = f_{\min}(n).$$

Since $\phi(M) = \phi(m) = n$,

$$\frac{M}{m} = \frac{M/\phi(M)}{m/\phi(m)}.$$

Because $m/\phi(m) \geq 1$, we have

$$\frac{M}{m} \leq \frac{M}{\phi(M)}.$$

Also, from the fibre-finiteness estimate above,

$$M \leq 2n^2 \leq 2x^2.$$

Therefore

$$\frac{f_{\max}(n)}{f_{\min}(n)} \leq \max_{1 \leq t \leq 2x^2} \frac{t}{\phi(t)} = (e^\gamma + o(1)) \log \log(2x^2) = (e^\gamma + o(1)) \log \log x.$$

Taking the maximum over all totient values $n \leq x$ gives

$$\mathcal{R}(x) \leq (e^\gamma + o(1)) \log \log x.$$

It remains to construct totient fibres with this order of separation. Let $y \rightarrow \infty$, and set

$$P_y := \prod_{p \leq y} p, \quad A_y := \prod_{p \leq y} (p-1).$$

By Linnik's theorem, choose a prime $\ell = \ell_y$ with

$$\ell \equiv 1 \pmod{A_y}, \quad \ell \leq CA_y^L.$$

Put

$$U_y := \frac{\ell - 1}{A_y}.$$

Let

$$Q_y := \prod_{\substack{q|U_y \\ q > y}} q,$$

the squarefree product of the prime divisors of U_y exceeding y . Define

$$a_y := \ell Q_y, \quad b_y := P_y U_y Q_y.$$

We claim that $\phi(a_y) = \phi(b_y)$.

Since $U_y < \ell$, the prime ℓ does not divide Q_y , and hence

$$\phi(a_y) = (\ell - 1) \prod_{\substack{q|U_y \\ q > y}} (q - 1) = A_y U_y \prod_{\substack{q|U_y \\ q > y}} (q - 1).$$

The prime divisors of b_y are exactly the primes $p \leq y$, together with the primes $q > y$ dividing U_y . Therefore

$$\begin{aligned} \phi(b_y) &= b_y \prod_{p \leq y} \left(1 - \frac{1}{p}\right) \prod_{\substack{q|U_y \\ q > y}} \left(1 - \frac{1}{q}\right) \\ &= P_y U_y Q_y \cdot \frac{A_y}{P_y} \cdot \prod_{\substack{q|U_y \\ q > y}} \frac{q-1}{q} \\ &= A_y U_y \prod_{\substack{q|U_y \\ q > y}} (q-1). \end{aligned}$$

Thus

$$\phi(a_y) = \phi(b_y) =: n_y.$$

Their ratio is

$$\frac{b_y}{a_y} = \frac{P_y U_y Q_y}{\ell Q_y} = \frac{P_y}{A_y} \cdot \frac{\ell - 1}{\ell}.$$

By Mertens' theorem,

$$\frac{P_y}{A_y} = \prod_{p \leq y} \frac{p}{p-1} = (e^\gamma + o(1)) \log y.$$

Also $\ell \rightarrow \infty$, so

$$\frac{\ell - 1}{\ell} = 1 + o(1).$$

Consequently

$$\frac{b_y}{a_y} = (e^\gamma + o(1)) \log y.$$

For large y , this ratio is greater than 1, and hence

$$\frac{f_{\max}(n_y)}{f_{\min}(n_y)} \geq \frac{b_y}{a_y} = (e^\gamma + o(1)) \log y.$$

It remains to ensure that $n_y \leq x$ for a suitable choice of y . Since $Q_y \leq U_y$,

$$n_y = \phi(a_y) = A_y U_y \prod_{\substack{q|U_y \\ q>y}} (q-1) \leq A_y U_y Q_y \leq A_y U_y^2.$$

Using $U_y = (\ell-1)/A_y$ and $\ell \leq C A_y^L$, we get

$$n_y \leq A_y \left(\frac{\ell-1}{A_y} \right)^2 \ll A_y^{2L-1}.$$

Moreover,

$$\log A_y = \sum_{p \leq y} \log(p-1) = \sum_{p \leq y} \log p + \sum_{p \leq y} \log \left(1 - \frac{1}{p} \right) = \vartheta(y) + o(y) = (1 + o(1))y.$$

Hence

$$\log n_y \leq (2L-1 + o(1))y.$$

Now choose

$$y = \frac{1}{4L} \log x.$$

Then

$$\log n_y \leq \left(\frac{2L-1}{4L} + o(1) \right) \log x < \log x$$

for all sufficiently large x . Thus $n_y \leq x$, and so

$$\mathcal{R}(x) \geq \frac{f_{\max}(n_y)}{f_{\min}(n_y)} \geq (e^\gamma + o(1)) \log y.$$

Since

$$\log y = \log \log x + O(1),$$

we conclude that

$$\mathcal{R}(x) \geq (e^\gamma + o(1)) \log \log x.$$

Together with the upper bound, this proves

$$\mathcal{R}(x) = (e^\gamma + o(1)) \log \log x.$$

□

3. A PERMANENCE OBSERVATION

The following elementary observation addresses the natural question of whether a single nontrivial totient collision forces infinitely many such collisions.

Proposition 3.1. *Suppose $a > b$ and $\phi(a) = \phi(b) = n$. Then there are infinitely many distinct totient values N such that*

$$\frac{f_{\max}(N)}{f_{\min}(N)} \geq \frac{a}{b}.$$

In particular, the existence of one nontrivial totient fibre implies the existence of infinitely many nontrivial totient fibres.

Proof. Choose any prime $r \nmid ab$. Since r is coprime to both a and b ,

$$\phi(ra) = (r-1)\phi(a) = (r-1)n$$

and

$$\phi(rb) = (r-1)\phi(b) = (r-1)n.$$

Thus the totient value

$$N_r := (r-1)n$$

has at least the two preimages ra and rb , and

$$\frac{ra}{rb} = \frac{a}{b}.$$

Therefore

$$\frac{f_{\max}(N_r)}{f_{\min}(N_r)} \geq \frac{a}{b}.$$

As r ranges over the infinitely many primes not dividing ab , the integers $N_r = (r-1)n$ are distinct. Hence infinitely many such totient values exist. $\square$